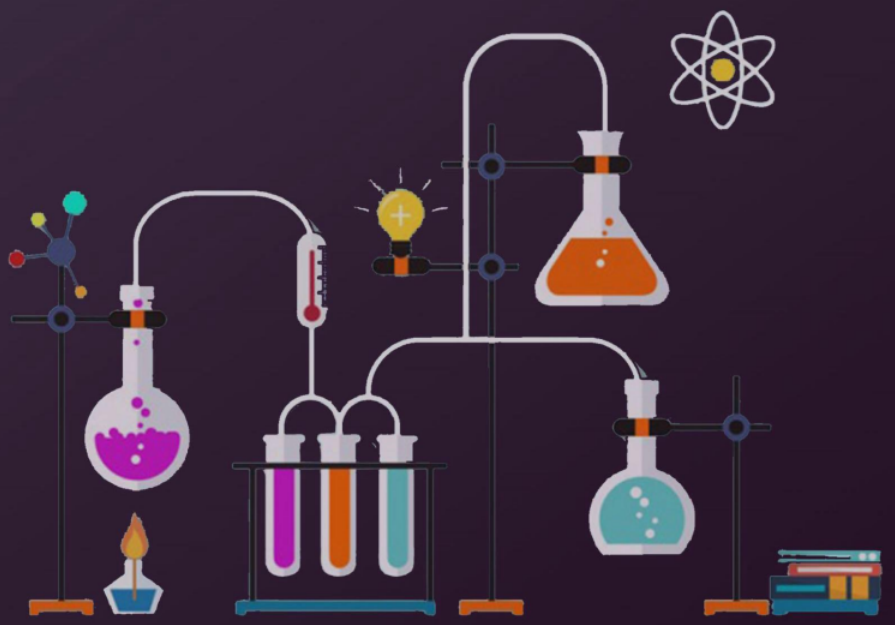
CHEMISTRY

Radioactivity



SHEET

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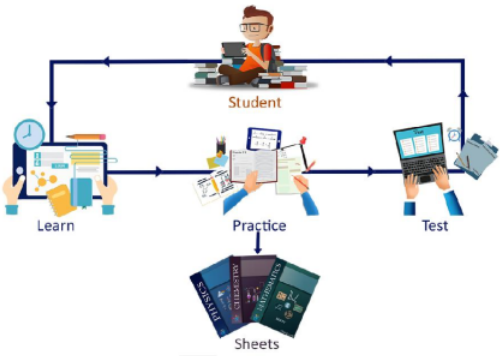
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

S.No	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Various types of Radioactive decay	1	1,4-5,9,12		1-4,15
2	Radioactivity Derivation	2		1	
3	Activity of a Radioactive sample	3-5	6-8,13	2, 5-21	
4	Application of Radioactivity	6-7	10,14	3	5-7
5	Nuclear fission reaction and Nuclear Reactor	8-9			
6	Nuclear Fusion	10			
7	ACC	11	2,3,11	4	8-14

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Various Type of Radioactive Decay

Q1. ${}^{14}_{6}C$ decays by emission of

(A) β^- (B) β^+ (C) n (D) α

Radioactivity Derivation

Q2. When an isotope undergoes K-capture, its mass number–

(A) Remains the same while the atomic number increases by one

(B) Remains the same while the atomic number increases by two

(C) Remains the same while the atomic number decreases by one

(D) Remains the same while the atomic number decreases by two

Activity of a Radioactivity Sample

Q3. Two radioactive material A_1 and A_2 have decay constants of 10 λ_0 and λ_0. If initially they have same number of nuclei, then after time $\frac{1}{9\lambda_0}$ the ratio of number of their undecayed nuclei will be

(A) $\frac{1}{e}$ (B) $\frac{1}{e^2}$ (C) $\frac{1}{e^3}$ (D) $\frac{\sqrt{e}}{1}$

Q4. The end product of (4n + 2) disintegration series the–

(A) ${}^{204}_{82}Pb$ (B) ${}^{208}_{82}Pb$

(C) ${}^{209}_{82}Pb$ (D) ${}^{206}_{82}Pb$

Q5. The radioactive sources A and B of half lives of t hr and 2t hr respectively, initially contain the same number of radioactive atoms. At the end of t hours, their rates of disintegration are in the ratio :

(A) $2\sqrt{2}$: 1 (B) 1 : 8

(C) $\sqrt{2}$: 1 (D) *l*n 2 : 1

Application of Radioactivity

Q6. The analysis of a mineral of uranium reveals that ratio of mole of ^{206}Pb and ^{238}U in sample is 0.2. If effective decay constant of process $^{238}U \longrightarrow {}^{206}Pb$ is λ then age of rock is

(A) $\frac{1}{\lambda}\ln\frac{5}{4}$ (B) $\frac{1}{\lambda}\ln\left(\frac{5}{1}\right)$

(C) $\frac{1}{\lambda}\ln\frac{4}{1}$ (D) $\frac{1}{\lambda}\ln\left(\frac{6}{5}\right)$

Q7. The half life of Tc^{99} is 6.0 hr. The delivery of a sample of Tc^{99} from the reactor to the nuclear medicine lab of a certain hospital takes 3.0 hr. What is the minimum amount of Tc^{99} that must be shipped in order for the lab to receive 10.0 mg?

(A) 20.0 mg (B) 15.0 mg

(C) 14.1 mg (D) 12.5 mg

Nuclear Fission Reaction and Nuclear Reactor

Q8. In the nuclear fission reaction a heavy nucleus splits up into two main fragments of lighter neclei and several ——

(A) Protons (B) Positrons

(C) Alpha particles (D) Neutrons

Q9. Nuclear reactors are based on–

(A) natural radioactivity

(B) nuclear fission

(C) nuclear fusion

(D) spontaneous chemical reaction

Nuclear Fussion

Q10. The reaction ${}_1^2D + {}_1^3T \rightarrow {}_2^4He + {}_0^1n$, is an example of–

(A) nuclear fission

(B) nuclear fusion

(C) artificial radioactivity

(D) radioactive disintegration

ACC

Q11. Choose the incorrect option-

(A) Among α, β and γ rays, γ ray has highest penetration power

(B) γ ray has highest velocity

(C) γ ray has highest ionization power

(D) α particle is also called helium nucleus.

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Select the correct statements

(A) A radioactive element decays by emitting one α and two β-particles. The daughter element formed is an isotope of the parent element.

(B) The daughter product formed by the emission of α-particle has mass number less by 4 units than the parent nuclide.

(C) ${}_{13}^{27}Al$ is a stable isotope hence ${}_{13}^{29}Al$ is expected to disintegrate by β-emission.

(D) Emission of a β-particle by a radioactive nuclide results in decrease in N / P ratio.

Q2. Select **correct** statement(s):

(A) The emission of gamma radiation involves transtition between energy levels within the nucleus.

(B) ${}_2^4He$ is formed due to emission of beta particle from tritium ${}_1^3H$.

(C) When positron (${}_{+1}^{0}e$) is emitted, $\frac{n}{p}$ ratio increases.

(D) Decay constant of radioactive substance is independent of temperature.

Q3. Radioactive disintegration rate is affected by-

(A) temperature (B) pressure

(C) electric field (D) time

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions.Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Questions 4 to 5

Mass defect in the nuclear reactions may be expressed in terms of the atomic masses of the parent and daughter neuclides in place of their nuclear masses.

Q4. The mass defect of nuclear reaction :

${}_4Be^{10} \rightarrow {}_5B^{10} + e^-$ is

(A) Δm = At. mass of ${}_4Be^{10}$ – At. mass of ${}_5B^{10}$

(B) Δm = At. mass of ${}_4Be^{10}$ – At. mass of ${}_5B^{10}$ – mass of one electron

(C) Δm = At. mass of ${}_4Be^{10}$ – At. mass of ${}_5B^{10}$ + mass of one electron

(D) Δm = At. mass of ${}_4Be^{10}$ – At. mass of ${}_5B^{10}$ – mass of two electrons

Q5. The mass defect of the nuclear reaction :

$_5B^8 \rightarrow {}_4Be^8 + e^+$ is

(A) Δm = At. mass of $_5B^8$ – At. mass of $_4Be^8$

(B) Δm = At. mass of $_5B^8$ – At. mass of $_4Be^8$ – mass of one electron

(C) Δm = At. mass of $_5B^8$ – At. mass of $_4Be^8$ + mass of one electron

(D) Δm = At. mass of $_5B^8$ – At. mass of $_4Be^8$ – mass of two electrons

Paragraph for Questions 6 to 8

Nuclei of a radioactive element 'A' are being produced at a constant rate, α. The element has a decay constant, λ. At time, t = 0, there are N_0 nuclei of the element.

Q6. The number of nuclei of A at time 't' is

(A) $\frac{\alpha}{\lambda}(1 - e^{-\lambda t})$

(B) $N_0 \cdot e^{-\lambda t}$

(C) $\frac{1}{\lambda}[\alpha - (\alpha - \lambda N_0)e^{-\lambda t}]$

(D) $\frac{N_0 \cdot \alpha}{\lambda}\left[1 - \left(1 - \frac{\lambda}{\alpha}\right)e^{-\lambda t}\right]$

Q7. If $\alpha = 2N_0\lambda$, the number of nuclei of A after one half life of A becomes

(A) zero (B) $2N_0$

(C) $1.5\ N_0$ (D) $0.5\ N_0$

Q8. If $\alpha = 4N_0\lambda$, the number of nuclei of A when A has reached steady state:

(A) zero (B) $4N_0$

(C) $2N_0$ (D) N_0

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q9.

Column-I	**Column-II**
(A) $n \longrightarrow p +$	(P) Positron emission
(B) $p^+ \longrightarrow n +$	(Q) β-emission
(C) X-ray emission	(R) K-electron capture
(D) $4[{}_1^1H] \longrightarrow$ .....$+ 2\beta^+ +$ energy	(S) α-emission

Q10.

Column-I	**Column-II**
(A) Nuclear fusion	(P) 4n series
(B) Carbon dating	(Q) (4n + 3) series
(C) Thorium series	(R) Age of dead
(D) Actinium series	(S) Hydrogen bomb

Q11.

Column-I	Column-II
(A) $^{2}_{1}D + ^{3}_{1}T \rightarrow ^{4}_{2}He + ^{1}_{0}n$ + energy	(P) b-emission
(B) $^{9}_{4}Be + ^{4}_{2}He \rightarrow ^{12}_{6}C + ^{1}_{0}n$	(Q) Artificial transmutation
(C) $^{24}_{12}Mg + ^{4}_{2}He \rightarrow ^{27}_{14}Si + ^{1}_{0}n$	(R) Discovery of neutrons
(D) $^{1}_{0}n \longrightarrow ^{1}_{1}H + ^{0}_{-1}e$	(S) Hydrogen bomb

SUBJECTIVE TYPE

Q12. Classify each of the following nuclides as "beta $\left(^{0}_{-1}\beta\right)$ emitter", or "positron $\left(^{0}_{+1}\beta\right)$ emitter": $^{49}_{20}Ca$ $^{195}_{80}Hg$ $^{8}_{5}B$ $^{150}_{67}Ho$ $^{30}_{13}Al$ $^{94}_{36}Kr$ Note: $^{84}_{36}Kr$ $^{200}_{80}$,Hg and $^{165}_{67}Ho$ are stable

Q13. Ac^{227} has a half life of 22 year w.r.t radioactive decay. The decay follows two parallel paths, one leading the Th^{227} and the other leading to Fr^{223}. the percentage yields of these two daughters nucleides are 2% and 98% respectively. What is the rate constant in yr^{-1}, for each of the separate paths?

Q14. The b count given by carbon obtained from an ancient fossil was 12 counts per second, while the b-count given by carbon sample from freshly cut tree was found to be 15 count per second. What is the age of fossil in years. Given : ($\lambda = 1.12 \times 10^{-4}\,yrs^{-1}$)

Exercise - 2 — *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The ratio of C^{14} to C^{12} in a living matter is measured to be $\frac{C^{14}}{C^{12}} = 1.3 \times 10^{-12}$ at the present time. Activity of 12.0 gm carbon sample is 180 dpm. The half life of C^{14} is nearly _____ $\times 10^{12}$ sec.

(A) 0.18 (B) 1.8

(C) 0.384 (D) 648

Q2. The radioactivity of a sample is R_1 at a time T_1 and R_2 at a time T_2. If the half life of the specimen is T, the number of atoms that have disintegrated in the time $(T_2 - T_1)$ is equal to-

(A) $(R_1T_1 - R_2T_2)$ (B) $(R_1 - R_2)$

(C) $(R_1 - R_2) / T$ (D)$(R_1 - R_2)T/0.693$

Q3. Wooden article and freshly cut tree show activity of 7.6 and 15.2 $min^{-1}\,gm^{-1}$ of carbon ($t_{1/2}$ = 5760 years) respectively. The age of article in years, is-

(A) 5760

(B) $5760 \times \left(\frac{15.2}{7.6}\right)$

(C) $5760 \times \left(\frac{7.6}{15.2}\right)$

(D) $5760 \times (15.2 - 7.6)$

Q4. A periodic table has 18 groups numbered from 1 to 18. What will be the group number of the final daughter nucleus formed, if $_{63}Eu^{150}$ shows sequential decay emitting 1α and 1β particles.

(A) 3 (B) 2

(C) 4 (D) 5

Q5. Radioactive sample (Z = 22) decreases 90% after 10 years. What will be the half life of sample.

(A) 5 years (B) 2 years

(C) 3 years (D) 10 years

Q6. Half life of radioactive ^{14}C is 5760 years. How many years will it take to reduce its sample of 200 mg to 25 mg :-

(A) 23040 years (B) 17280 years

(C) 11520 years (D) 5760 years

Q7. The half-life of a radio isotope is four hours. If the initial mass of the isotope was 200 g, the mass remaining after 24 hours undecayed is :

(A) 3.125 g (B) 2.084 g

(C) 1.042 g (D) 4.167 g

Q8. For an alpha-emitting isotope. The value of the disintegration constant is 0.49×10^{-10} per year. The amount of the isotope of a given sample will reduce to half its value after a period [in years] of nearly :-

(A) 0.45×10^{10} (B) 0.9×10^{10}

(C) 1.4×10^{10} (D) 2.8×10^{10}

Q9. If half lives of a radioactive element for undergoing parallel path α-decay and β-decay are 4 years & 12 years respectively. Percentage of element that remains after 12 years will be :

(A) 50% (B) 12.5%

(C) 6.25% (D) 25%

Q10. If 50 sec. required to decay 2/3rd fraction of a radio active sample. What fraction will be left after 25 sec.

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{3}}$

(C) $\frac{1}{3}$ (D) $\sqrt{\frac{2}{3}}$

Q11. For the nuclear reaction ;

$_{Z}^{A}X \longrightarrow {}_{Z-4}^{A-8}Y$; $t_{1/2}$ = 1600 yrs. If initial activity was 10^7 dps, how many α-particles will be emitted per second after 4800 years ?

(A) $1.25 \times 10^6\ s^{-1}$ (B) $2.5 \times 10^6\ s^{-1}$

(C) $1.25 \times 10^7\ s^{-1}$ (D) None

Q12. If the amount of radioactive substance is increased three times and simultaneously temperature also increases thrice, the number of atoms disintegrating per unit time would be -

(A) $\frac{1}{3}^{rd}$ of original atoms

(B) constant

(C) Triple

(D) 9 time

Q13. Number of natural life times (T_{av}) required for a first-order reaction to achieve 99.9% level of completion is :

(A) 2.3 (B) 6.9

(C) 9.2 (D) 0.105

Q14. If time t is required for a radioactive substance to become one third of it's initial amount, what fraction would be left after 0.5 t :-

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{3}}$

(C) $\frac{1}{3}$ (D) $\sqrt{\frac{2}{3}}$

Q15. The half life of a radioactive element is 20 min. The time interval between the stages of its 33% and 67% decay is :-

(A) 40 min (B) 20 min

(C) 30 min (D) 25 min

Q16. 80 % the radioactive nucleii are left undecayed in one day then % of nucleii left undecayed after two days will be -

(A) 40% (B) 60%

(C) 64% (D) 46%

Q17. Two radioactive nuclides A and B have half life 175 min and 25 mins respectively. A fresh sample contains nuclides of B to be 32 times that of A. How much time should elapse so that number of nuclides of A becomes twice that of B ?

(A) 100 min. (B) 150 min.

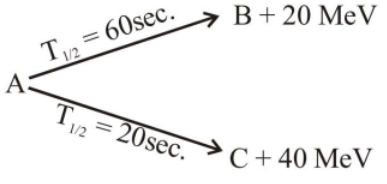
(C) 175 min. (D) never

Q18. Gadolinium – 153, which is used to detect osteoporosis (porous bones), has a half life of 242 days. Which value is closest to the percentage of Gd– 153 left in a patient's system after 2 years -

(A) 33 % (B) 26 %

(C) 12.0 % (D) 6.25 %

Q19. In given parallel radioactive decay find energy liberated per atom of 'A' in MeV.



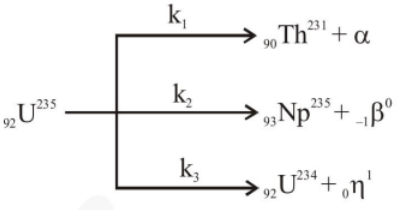
(A) 35 (B) 25

(C) 60 (D) 30

Q20. For the radioactive parallel decay After 30 min. from the start of decay the mole ratio of products obtained are :

$Th^{231} : Np^{235} : U^{234} :: 4 : 2 : 1.$

If $k_2 = 2 \times 10^{-2}$ sec^{-1} then the overall decay constant of $_{92}U^{235}$ is -



(A) 3×10^{-2} sec^{-1}

(B) 8×10^{-2} sec^{-1}

(C) 2×10^{-2} sec^{-1}

(D) 7×10^{-2} sec^{-1}

Q21. If 10^{23} atoms added after each half life in a sample of 10^{24} atoms, then number of atoms disintegrated between 2nd and 3rd half life is

(A) 2×10^{23}

(B) 1.5×10^{23}

(C) 10^{23}

(D) 4×10^{23}

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

PARAGRAPH TYPE

This section contains ***PARAGRAPHS****. Based on each paragraph, there are questions. Each question has FOUR options.* ***ONLY ONE*** *of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph For Questions 1 to 4

In the disintegration of a radioactive element, α and β-particles are evolved from thenucleus.

$${}^{1}_{0}n \longrightarrow {}^{1}_{1}H + {}^{0}_{-1}e + \text{antineutrino} + \text{Energy}$$

$$4\,H \longrightarrow He + 2e + \text{Energy}$$

Then, emission of these particles changes the nuclear configuration and results into a daughter nuclide. Emission of α-particle results into a daughter element having atomic number lowered by 2 and mass number by 4. On the other hand, emission of a β-particle yields an element having atomic number raised by one.

Q1. Which of the following combinations give finally an isotope of the parent element ?

(A) α,α,β (B) α,γ,α

(C) α,β,β (D) β,γ,α

Q2. A radioactive element belongs to III (B) group ; it emits one α and one β-particle to form a daughter nucleide. The position of daughter nuclide will be in

(A) II A (B) I A

(C) II B (D) IV B

Q3. How many α and β particles would be emitted from a nuclide so that an isobar is produced ?

(A) 1α, 1β (B) 1α, 2β

(C) 2α, 2β (D) 4β

Q4. Which of the following statement (s) is/ are correct ?

(A) Emission of a β-particle results in isodiaphere of parent element

(B) Emission of one α and two β-particles results in isotope of the parent element

(C) Emission of γ-radiations may yield nuclear isomer

(D) All of the above are correct.

Paragraph For Questions 5 to 7

Carbon 14 is used to determine the age of orgainc material. The procedure is based on the formation of ^{14}C by neutron capture in the upper atmosphere.

$${}_{7}N^{14} + {}_{0}n^{1} \longrightarrow {}_{6}C^{14} + {}_{1}H^{1}$$

^{14}C is absorbed by living organisms during photosynthesis. The ^{14}C content is constant in living organism once the plant or animal dies, the uptake of carbon dioxide by it ceases and the level of ^{14}C in the dead being falls due to the decay which C^{14} undergoes.

$${}_{6}C^{14} \longrightarrow {}_{7}N^{14} + {}_{-1}e^{\circ}$$

The half life period of ^{14}C is 5770 years. The decay constant (λ) can be calculated by using the following formula $\lambda = \frac{0.693}{t_{1/2}}$

The comparison of the β^- activity of the dead matter with that of carbon still in circulation enables measurement of the period of the isolation of the material from the living cycle. The method however, ceases to be accurate over periods longer than 30,000 years. The proportion of ^{14}C to ^{12}C in living matter is $1 : 10^{12}$

Q5. Which of the following option is correct ?

(A) In living organisms, circulation of ^{14}C from atmosphere is high so the carbon content is constant in organism

(B) Carbon dating can be used to find out the age of earth crust and rocks

(C) Radioactive absorption due to cosmic radiation is equal to the rate of radioactive decay, hence the carbon content remains constant in living organism

(D) Carbon dating cannot be used to determine concentration of ^{14}C in dead beings.

Q6. What should be the age of fossil for meaningful determination of its age?

(A) 6 years

(B) 6000 years

(C) 60000 years

(D) it can be used to calculate any age

Q7. A nuclear explosion has taken place leading to increase in concentration of C^{14} in nearby areas. C^{14} concentration is C_1 in nearby areas and C_2 in areas far away. If the age of the fossil is determined to be t_1 and t_2 at the places respectively, then

(A) The age of the fossil will increase at the place where explosion has taken place and $t_1 - t_2 = \frac{1}{\lambda} \ln \frac{C_1}{C_2}$.

(B) The age of the fossil will decrease at the place where explosion has taken place and $t_1 - t_2 = \frac{1}{\lambda} \ln \frac{C_1}{C_2}$.

(C) The age of fossil will be determined to be the same.

(D) $\frac{t_1}{t_2} = \frac{C_1}{C_2}$.

Paragraph For Questions 8 to 11

There are four radioactive decay series called thorium (4n) ; uranium (4n + 2) ; actinium (4n + 3) and neptunium (4n + 1) series. Neptunium series is artificial while other three series are natural. End products of each radioactive decay series have stable nuclei. All natural decay series terminate at lead but neptunium or artificial series terminate at bismuth.

Q8. The end product formed in the disintegration of $^{222}_{88}Ra$ is -

(A) $^{204}_{81}Tl$ (B) $^{206}_{82}Pb$

(C) $^{222}_{86}Rn$ (D) $^{207}_{83}Bi$

Q9. Rn is a member of actinium series. Another member of same series is.

(A) U (B) Ac

(C) Th (D) Po

Q10. The end products of uranium and actinium series are respectively.

(A) Pb – 206 ; Pb – 207

(B) Pb – 206 ; Pb – 208

(C) Pb – 207 ; Pb – 208

(D) Pb – 206 ; Bi – 208

Q11. The starting isotope and the end product isotope of actinium series are :

(A) Ac and Pb

(B) U and Pb

(C) U and Ac

(D) None

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q12.

Column-I	Column-II
(A) Disintegration constant	**(P)** 1.44 $t_{1/2}$
(B) Average life	**(Q)** 1 dps
(C) One curie	**(R)** $\frac{0.693}{t_{1/2}}$
(D) One Becquerel	**(S)** 3.7×10^{10} dps

TRIPLE MATCH

*This section contains TWO tables (each having 3 columns and 4 rows). Based on each table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Column-I	Column-II	Column-III
(I) Isotones	(i) Same mass number	(P) $^{234}_{91}$Pa and $^{234}_{90}$Th
(II) Isobars	(ii) Same neutrons	(Q) $^{12}_{6}$C and $^{14}_{6}$C
(III) Isotopes	(iii) Same difference between Neutron & proton	(R) $^{39}_{19}$K and $^{19}_{9}$F
(IV) Isodiaphers	(iv) Same atomic number	(S) $^{39}_{18}$Ar and $^{40}_{19}$K

Q13. Select the correct match –

(A) (I), (ii), (R) (B) (II), (i), (P) (C) (III), (ii), (Q) (D) (IV), (iv), (R)

Q14. Select the incorrect match –

(A) (I), (ii), (S) (B) (II), (i), (P) (C) (III), (iv), (Q) (D) (IV), (iii), (R)

NUMERICAL TYPE

Q15. Complete the following nuclear equations:

(A) $^{14}_{7}N + ^{4}_{2}He \rightarrow ^{17}_{8}O + ...$

(B) $^{9}_{4}Be + ^{4}_{2}He \rightarrow ^{12}_{6}C + ...$

(C) $^{9}_{4}Be\ (p, a)..... x$

(D) $^{30}_{15}P \rightarrow ^{30}_{14}Si + ...$

(E) $^{3}_{1}H \rightarrow ^{3}_{2}He + ...$

(F) $^{43}_{20}Ca(\alpha,....) \rightarrow ^{46}_{21}Sc$

(G) $^{23}_{11}Na + ^{4}_{2}He \rightarrow ^{26}_{12}Mg + ...$

(H) $^{64}_{29}Cu \rightarrow \beta^{+} + ...$

(I) $^{106}Ag \rightarrow ^{106}Cd + ...$

(J) $^{10}_{5}B + ^{4}_{2}He \rightarrow ^{13}_{7}N + ...$

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
B	C	A	D	C	D	C	D	B	B
11									
C									

Ex-1A

1. A,B,C,D **2.** A,C,D **3.** D **4.** A **5.** D

6. C **7.** C **8.** B

9. (A) → Q; (B) → P; (C) → R; (D) → S **10.** A → S; B → R; C → P; D → Q

11. (A) → S ; (B) → Q, R; (C) → Q; (D) → P

12. beta emitter : ^{49}Ca, ^{30}Al, ^{94}Kr, positron emitter : ^{195}Hg, ^{8}B, ^{150}Ho

13. ($6.30\times10^{-4}\,yr^{-1}$, $3.087\times10^{-2}\,yr^{-1}$) **14.** (1992.7)

Ex-2									
1	2	3	4	5	6	7	8	9	10
A	D	A	A	C	B	A	C	C	B
11	13	14	15	16	17	18	19	20	21
B	C	B	B	B	C	C	C	A	D
22									
A									

Ex-2A

1. C **2.** A **3.** D **4.** D **5.** C

6. B **7.** A **8.** B **9.** A **10.** A

11. B **12.** A → R; B → P; C → S; D → Q **13.** B

14. D

15. (A) $^{1}_{1}H$, (B) $^{1}_{0}n$, (C) $^{6}_{3}Li$, (D) $^{0}_{+1}e$, (E) $^{0}_{-1}e$, (F) p (proton),

(G) $_{1}H^{1}$, (H) $_{28}Ni^{64}$, (I) b^{+}, (J) $_{0}n^{1}$